

Women Safety with Location Tracking & Emergency Alerts

Dr. T. Venkata Naga Jayudu
M. Tech, Ph.D
*Dept. of Computer Science and
Engineering
Srinivasa Ramanujan Institute
of Technology*

G Nandini
*Dept. of Computer Science and
Engineering
Srinivasa Ramanujan Institute
of Technology
Ananthapuram, India*

M Sukanya
*Dept. of Computer Science and
Engineering
Srinivasa Ramanujan Institute
of Technology
Ananthapuram, India*

V Akhila
*Dept. of Computer Science and
Engineering
Srinivasa Ramanujan Institute
of Technology
Ananthapuram, India*

Abstract-- To secure women in the emergency case, it is vital to obtain reliable mechanisms working in real-time and operating under the high stress and in the limited space of conditions. This project offers a Women Safety System that combines SOS activation on the mobile level, with the real-time tracking of the GPS location, and a centralized web monitoring station. When activated by an emergency alert, the system automatically records and sends accurate location information to emergency contacts that have been registered with the system, so that live tracking of their position can be continued and immediate situational awareness can be obtained until the alert goes off. The architecture that has been proposed concentrates on low-latency communication, usability, and operational reliability thus lowering the response time and the need to resort to manual intervention. Once the system has been practically validated in the course of an experiment, it becomes possible to declare that the system is practically feasible and effective and can provide a scalable and expandable platform to the development of intelligent, technology-based, women-focused solutions to safety in smart environments.

Keywords/Phrases Women Safety System, SOS Alert Mechanism, Real-Time GPS Tracking, Live Location Sharing, Emergency Calling, Flutter Mobile Application, Centralized Monitoring, Emergency Response Systems.

I. INTRODUCTION

The safety of women is one of the important real-life issues, especially in urban and semi-urban areas where cases of harassment, stalking, and physical violence have become a common occurrence

during their daily activities that include commuting to work, working late hours, or traveling alone. In emergency circumstances, the difficulty is sometimes not the deficiency of a communication infrastructure, but the incapacity of people to successfully apply it in a condition of stress, intimidation or bodily limitation. Empirical research and reports all over the world show that response to emergency and lack of location awareness contribute largely and severely to the severity of such a situation [1]. Thus, the old safety systems based on voice calls and manual reporting cannot be sufficient in situations with a time constraint.

The advent of smartphones and GPS enabled service has resulted in the creation of a number of mobile based safety apps designed to ease emergency communication [2]. Despite the fact that such systems show the possibility of technology-based safety support, much of the current design can be seen to have weaknesses when subjected to the dynamics of real-life emergencies. Some of the most common are reliance on repeated user interaction, no persistent location tracking when an alert has been initiated and no centralized monitoring facilities where the responders or trusted contacts can keep continual situational awareness [3]. These limitations lower the effectiveness of the system especially when the emergency is long-lasting or there is movement of the user.

Also, the problem of the latency of communication, system scalability, and operational reliability in the limited network conditions is not addressed fully [4]. The absence of unified web-based systems to support live tracking, alert management and post

incident analysis further restrains the coordinated response and accountability [5]. These loopholes underline the necessity to have a coherent safety framework that aligns technological expertise and realistic emergency demands.

In a bid to overcome these difficulties, this paper comes up with a Women Safety System that combines instant SOS activation system with real-time GPS location tracking and the web-based monitoring interface being centralized [6]. The system is created to reduce the effort of users, offer complete location visibility until alert termination, and coordinate emergency response better.

This rest of the paper is structured as follows. Section II discusses related literature on the issue of women safety systems and emergency alerts technologies. Section III explains the system design and the methodology proposed. IV gives the experimental setup, results and analysis. Section V is the discussion of the findings and their practical implications, and the last section of the paper, Section VI, is a conclusion of the research and its future directions.

II. RELATED WORK

The most recent research on the topic of women safety systems focuses on the fact that the gap between actual emergency situations in the real world and the extent of the technological solutions

available is widening. Most of the incidents are common in late-night travelling, in isolated locations.

commuting commutes, or circumstances of constant movement, in which quick reaction and constant awareness of situations is crucial. This motivation is frequently justified by previous studies with the help of statistical data and comparative analysis through visual image, proving that free or delayed reaction to emergencies will severely decrease the possibility of timely response.

A. Women Safety Applications on the mobile.

The main idea of the early mobile-based women safety apps was that people could activate emergency alerts with the least amount of interaction. Sharma et al. [1] suggested a safety system on mobile phones, which sends an alert message and the location information when a user triggers the system. Such systems are normally drawn in application flow diagrams and interface screen shots in real world tests to show the behavior of alert initiation. But according to real-life application analysis, it is observed that such applications usually relay the location of the user at the time of alert activation only. Conceptually as depicted in Fig. 1, after the user initiates movement after receiving the alert, no other positional updates are sent to the responders making it ineffective in dynamic situations.

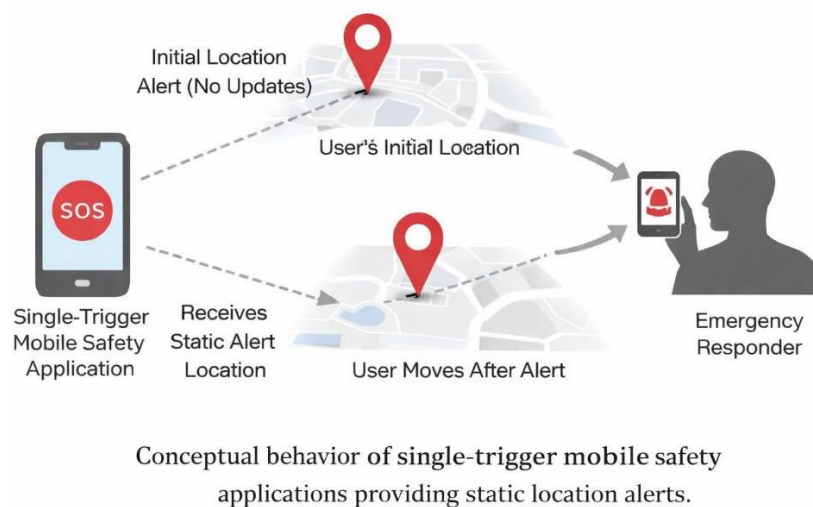


Fig. 1. Theoretical behavior of single-trigger mobile safety applications that give static location alarms.

B. GPS-Enabled Emergency Tracking Systems

To address the constraints of a non-moving alert, a number of scholars have decided to investigate

continuous GPS-based tracking systems. Rahman and Hossain [7] showed that it is very important that periodic transmission of location updates leads to high accuracy of the responders in moving scenarios. Related studies typically conduct

performance evaluation with the aid of tracking-path visualization and update interval graph to examine system behavior when mobility is a factor. Continuous tracking enables the responders to track the movement path of a user as illustrated in

Fig. 2 instead of having a fixed point. However, most of such systems are aimed at monitoring accuracy and do not consider the duration of the tracking or the management of emergency states after they have been triggered.

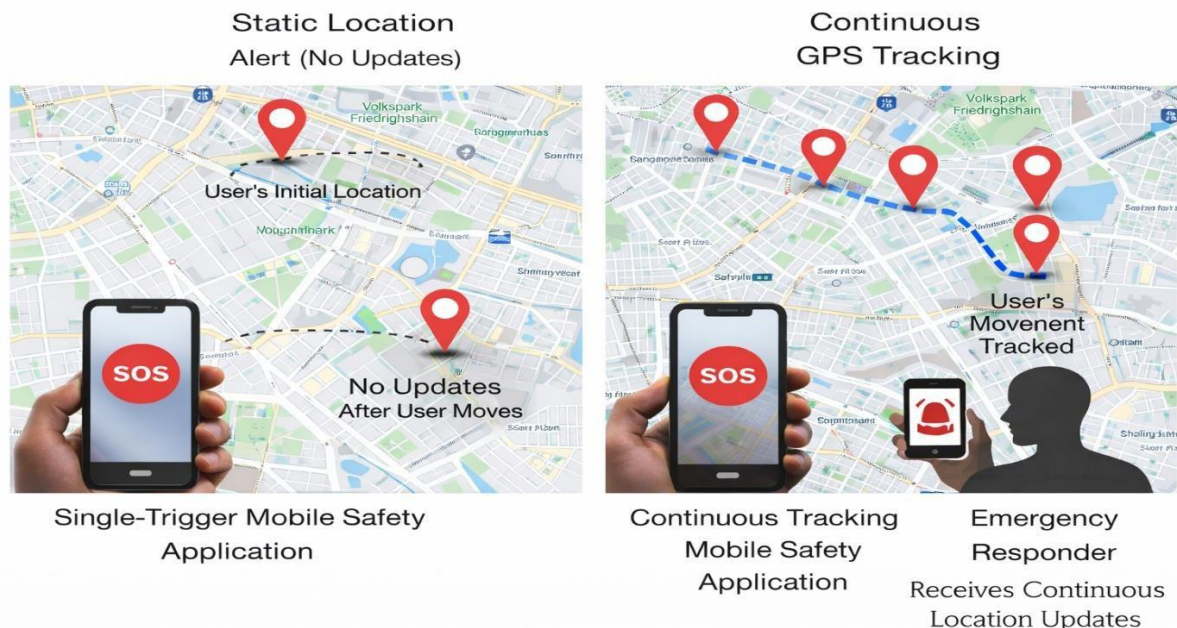


Fig. 2. Comparison of fixed location alerts and continuous movement tracking by a GPS position.

C. Centralized Monitoring and Visualization Platforms

Current developments have come up with centralized monitoring systems in an effort to enhance coordination in emergency response efforts. Mehta and Joshi [8] introduced a web-based dashboard, visualizing live alerts information, location information, with interface screenshots and analysis of response time. Such

dashboards allow the responders to control a variety of emergencies at the same time. The centralized visualization, as shown in Fig. 3, enhances the situational awareness by consolidating the alert status data and location information as they happen. Nonetheless, implementations available in use tend to be independent of mobile-side alert logic resulting in disjointed emergency management and does not support much history and alert lifecycle control.

Emergency health services dashboard with wait time

This slide highlights how an emergency health services dashboard, featuring waittime data, improves decision-making, optimizes resources, and enhances patient care in real time for efficient emergency department management.



Fig. 3. Existing emergency response systems have typical centralized monitoring dashes.

D. Observed Limitations and Research Motivation

In a comparative analysis of previous work with the assistance of application workflows, tracking visualization and dashboard interfaces, constant gaps are identified. They are the absence of a smooth transition between the alert generation and active monitoring, and the irregular management of the emergency lifecycle and communication delays in the case of a limited network bandwidth [6], [9]. The visuals provided in Figs. 1-3 all point to the fact that there is no unified, end-to-end solution to safety existing in systems. These findings are the direct inspiration of the combined system suggested in this paper that is aimed at the sustained tracking, centralized monitoring, and logical experimental evaluation based on the graphs of response-time, the tracking trajectories, and interface screenshots.

III. PROPOSED WORK

The proposed work is a Women Safety Application built on the Flutter platform and aimed at helping users obtain quick help in case of an emergency situation with the help of the instant activation of SOS, constant GPS monitoring, and real-time location sharing. Real-life examples are an incentive behind the system because victims might not be able to make a call, write a message, or manually share where they are. To solve this, the suggested application provides an opportunity to trigger the case of an emergency with a single touch and automatically share the live position of the user with his trusted people and the authorities concerned. The solution will include a mobile app to launch an SOS and a monitoring interface that can visualize the live location updates to assist in timely intervention.

A. Key Functionalities

The following are the main functionalities contained in the proposed system:

- **SOS In the event of an emergency:** The activation is one-tapped to have an emergency session and inform registered contacts.
- **Real-Time GPS Tracking:** Continuous gathering of gps positions following SOS activation to keep track of the situation during the duration of the emergency.
- **Location Sharing via Live Link:** Automatic generation and dissemination of a live tracking link/URL to allow the contacts to see the current position of the user with the help of standard map services.

- **Emergency Calling Feature:** Feature. call out capability to predefined contacts or complementary emergency services. Multi-based alerts which are message-based channel communication.
- **Emergency Contact Management:** Confidential storage and maintenance of trusted used contacts (family/friends/authorities) during emergencies.
- **Centralized Monitoring Support:** A responder-side interface which shows live. following data to better coordination. and response.
- **Cancellation of Alerts:** Controlled dissolution of the crisis meeting once the user does not get into trouble, and there is no needless escalation.
- **History and Audit Trail:** Storage of sessions of emergency (timestamps and monitoring environment) in case of future examination and accountability

B. Benefits of the Proposed System.

The advantages of the proposed work in relation to traditional safety applications are the following:

1. Less than the ideal interaction with distress.
2. with a one-touch SOS button.
3. Constant monitoring and not fixed point notification.
4. Location intelligence of actionable data based on live maps links that provide real-time navigation to responders.
5. Multi-channel emergency communications via dissemination of alerts and calling support.
6. Better coordination by centralized monitor and real time visualization.
7. Control and responsibility in operation through cancellation of alerts and maintenance of history.

C. Modules of the Proposed System

The implementation aligned modules are structured into the system:

1. User Interface Module (Flutter): It also offers home access, SOS trigger, contact management, and emergency controls.
2. SOS & Notification Module: The emergency session is initiated and alerts sent to registered contacts.
3. GPS Tracking Module: This constantly records updates of the position at a specified rate.
4. Location Sharing Module: Turns GPS in

URL/live tracking links of a map and spreads them automatically.

5. **Emergency Calling Module:** Provides the ability to call emergency contacts/services directly and communicate using voice in emergency.
6. **Monitoring Module:** Shows real-time location updates in a monitor interface among the responders or the authorized watchers.
7. **Lifecycle/History Module:** Deals with the notification of the termination of alerts, and archives emergency session logs.

D. Working of the Proposed System

Upon the user using SOS feature, the application takes immediate GPS position and initiates an emergency session. The system then sends an alert to the registered contacts with a live tracking link to the position of the user. At the same time, the process of constant tracking starts and regularly displays the location of the user to enable the responders to track the movement in real time. Simultaneously, the user may also use the calling feature so as to be able to directly contact the emergency services or contacts and to be able to communicate even in case of insufficient messaging. The emergency session is completed till the user can cancel the alert and then the system stops tracking, records session information in the history file to be referred to in the future.

IV. IMPLEMENTATION

The proposed Women Safety System is adopted as a mobile based application which is aimed at facilitating emergency alerting, constant location tracking, live location sharing, and emergency calling. The application is built with Flutter, which allows becoming cross-platform and provides a responsive user interface that can be applied to the real-life emergency situation. Its implementation is centered on consistent execution, the presence of minimum user communication and integration of the system elements. Upon opening the application, the user will be presented with the home interface that will give him direct access to the functionalities of safety-related applications. Users in the first setup make known trusted emergency contacts, including family members, friends or local authorities. Such contact information is constantly maintained and automatically used in cases of emergency to keep people ready beforehand and so that it does not require human intervention in times of distress.

When SOS is activated, the application then takes the current GPS position of the user and opens an emergency session. The location information taken is processed and sent to the registered emergency contacts in various forms, such as geographic coordinates and map-based live location URLs, so that the contacts of the user can see the location of the user on a standard mapping service. This is only carried out in an event where location services and network connection is in place which guarantees a reliable passage of data.

Once the emergency session is started, the application will go into constant tracking. In this mode, new GPS coordinates are taken and sent every now and then which enables users to be tracked in real time. This tracking system will provide the emergency contacts and responders with the latest location details as opposed to using a single alert which is inactive. Simultaneously, the application also has a calling feature, allowing the user to make the calls directly to predefined emergency contacts/services, thus facilitating the multi-channel emergency contact/service.

An integrated architecture facilitating the implementation of the proposed system is the coordination of mobile interaction, backend communication and monitoring services. Fig. 4 displays the architecture diagram that illustrates the relationship that exists between the Flutter-based mobile application, GPS services, backend server and the responder-side monitoring interface. The backend component provides SOS session state management, location update synchrony, alerts distribution, and emergency history. This architecture design provides a consistent behavior of the systems and real time synchronization across the emergency lifecycle.

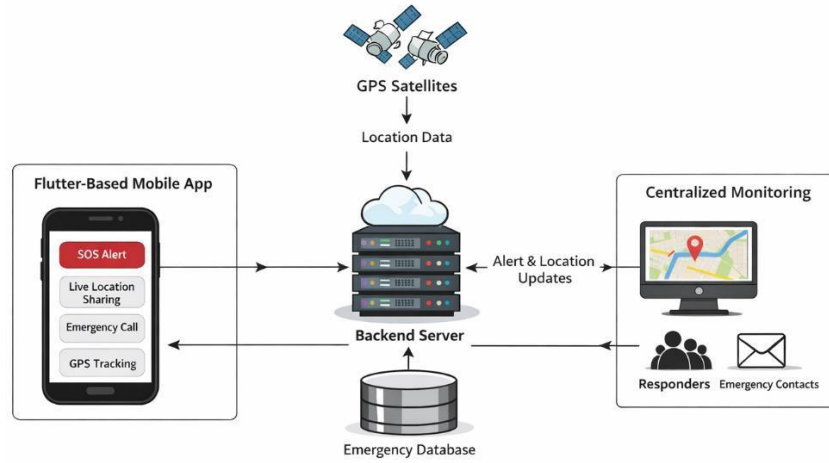


Fig. 4. The Architecture diagram of the proposed Women Safety System.

The emergency session can be canceled by use of the cancel SOS feature once the user is in a safe condition. This measure prevents location streaming, synchronizes the emergency status of all the components, and further prevents the dissemination of alerts. Emergency sessions done are stored in the system history, and this allows a review and accountability of the incident done.

In general, the implementation shows that the Women Safety System suggested to use Flutter is effective to combine SOS alerting, constant tracking of position, sharing of location (live), making calls in emergencies, and centralized coordination. The implementation process is architecture-based to guarantee consistency of performance in the case of real-life emergency response and is the foundation of the experiment described in the following section.

V. RESULTS

This section is a report about the findings acquired once the proposed application Women Safety has been implemented and tested. Just like the base paper, the findings are presented in the form of application screenshots taken when running the code. This section is aimed at showing how the application works and proving its applicability to the real-life emergency situation.

Once the application starts, the user gets the central interface with access to all functions relating to safety. This display informs the user that the

application is ready to use and enables the user to take emergency actions without his/her intricate navigation. Fig. 5 presents the home screen of the application that serves as the point of entry of all emergency operations.

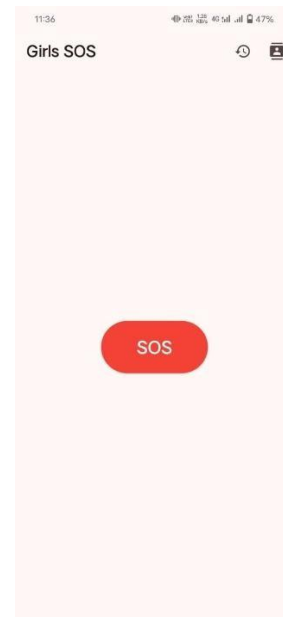


Fig. 5. Women Safety Application Home screen

Once the application is initialized, the user can After the application is launched, the user is able to program emergency contact information in advance. Such contacts are people who are trusted to include family members, friends, authorities to whom alerts will be sent in case of emergency. As Fig. 6 demonstrates, the emergency contact configuration screen confirms that the application is able to store and manage registered contact information.

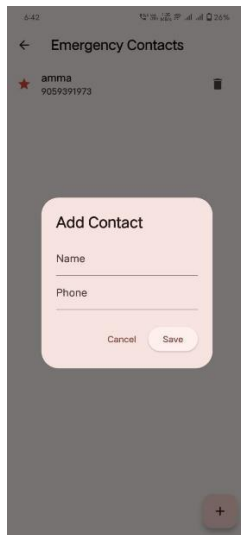


Fig. 6. Emergency contact arrangement screen

In case of an emergency, the user can trigger the SOS entity by pressing the SOS button that is located within the application interface. When activated, the application instantly captures the location of the user in their current GPS position and sends out the emergency signal. Fig. 7 demonstrates the SOS activation screen, which proves the fact that the alert may be triggered with only one interaction, and it is very important under distress conditions.

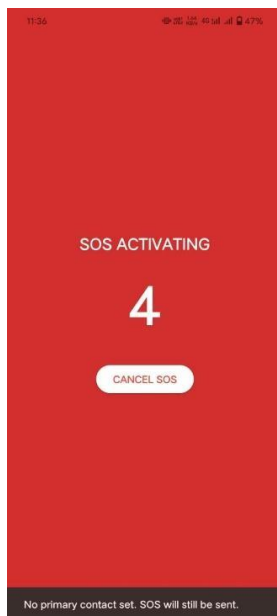


Fig. 7. SOS activation screen

Once the SOS alert has gone off, the application begins exchanging the location of the user with emergency contacts registered. The geographic positioning is distributed as a live tracking link that may be viewed with the help of regular map service providers. The location sharing interface is presented in Fig. 8, and it proves that the application is able to create and send real-time location information.

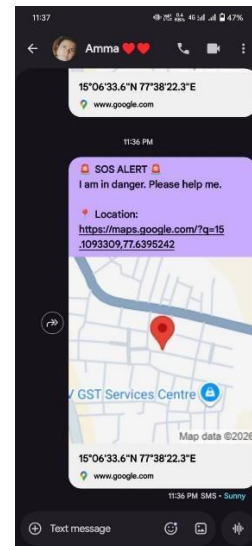


Fig. 8. Location sharing interface with live track connection.

Continuous real-time tracking of location can also be done on the application once SOS is activated. As the user is on the move, current location data are periodically sent which gives emergency contacts the opportunity to track movement in their dynamically changing location. The live tracking behavior is shown in Fig. 9 with several location updates being shown, creating an movement path as opposed to a single location.

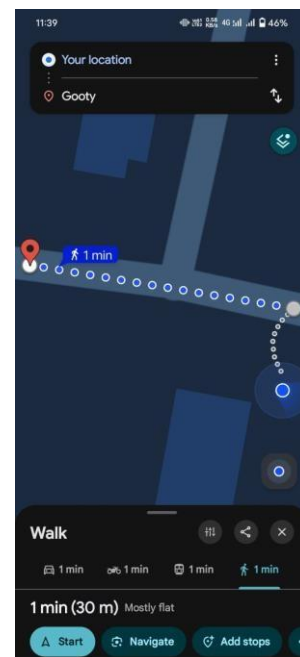


Fig. 9. Live location tracking monitor.

The application also has a calling service that the user can use to contact emergency contacts or services directly, besides alerts that are in form of messages. This feature will make voice communication possible even in times of emergency. Fig. 10 displays the calling interface, which proves the successful implementation of the emergency calling feature.

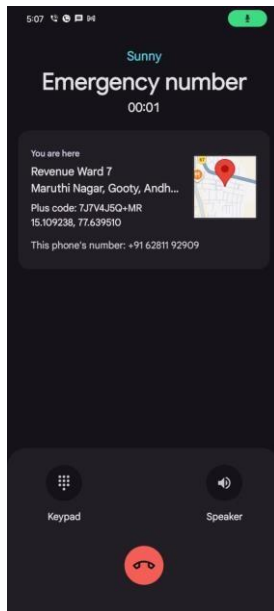


Fig. 10. Interface of emergency calling.

Upon getting to a safe state; the emergency alert may be canceled through the cancel SOS option. When canceled, the application will halt location tracking and (locally) update the emergency status. Fig. 11 displays the cancel SOS screen where termination of the emergency session is controlled.

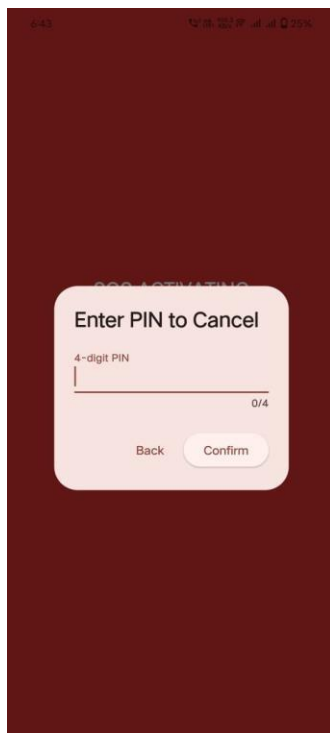


Fig. 11. Cancel SOS screen

Another feature of the application is the record of the emergency sessions completed. This history enables the users to view the history of the alerts as well as the details of the alerts like time and place. Fig. 12 indicates that the emergency history screen is displayed, which proves that the system is able to store and display previous emergency records.

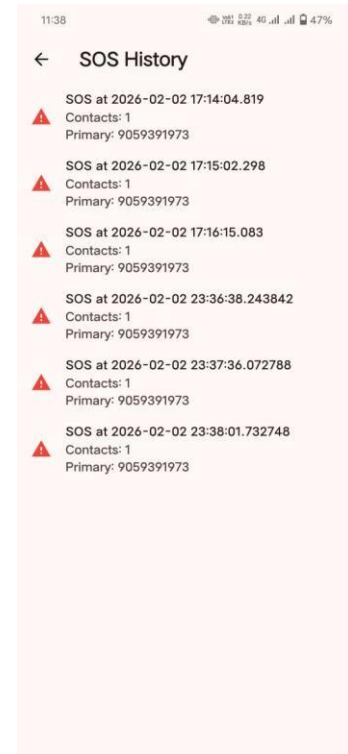


Fig. 12. Emergency history screen

Based on the results obtained, it can be concluded that, the Women Safety Application proposed works well in all the functionalities expected. The screenshots confirm the instant SOS activation, real time sharing of location, nonstop tracking, emergency calling and lifecycle management. These findings validate that the system is reliable and can be used in practice in an emergency.

VI. DISCUSSION

The findings made in the above section show that the proposed Women Safety Application successfully overcomes some of the most significant weaknesses noted in the existing kinds of safety systems. These results are discussed in relation to how these findings relate to extreme real-world emergency situations, and how the suggested design options can be used in enhancing the usability, reliability, and responsiveness.

Among the most important findings, the use of a single-action SOS activation in parallel with location tracking should be mentioned. In real life moments of distress, users can develop panic, physical restriction or lack of time to interact with their devices. Being able to send out an emergency notification with minimal level of interaction makes delay in response significantly shorter and the chances of timely response much higher. The findings validate the use of the application that does not activate an alert repeatedly and thus complies with the practicality of emergency needs.

Continuous GPS-based tracking is a significant advancement over the old traditional systems of static alerts. Continuous updates are unlike one-time location sharing and they offer dynamic situational awareness to the responders especially where there is user movement. This is important in city and semi urban setting where the victims might be on transit. The tracking behavior observed proves the fact that the system provides the right and current location visibility during the emergency period.

The other noticeable point that was suggested by the results is the importance of live sharing of locations by using map- based URLs. The system provides easily understandable and easily usable information on locations through the sharing of the locations in a format that can be easily read and used by the emergency contacts without the use of specific tools like the widely used mapping services. This design option will make it easier to access and use, particularly by non-technical users.

The system is also highly reinforced by the fact that it has an emergency calling feature which allows multi-channel communication. Whereas messages and sharing of locations prove to be useful, voice communication is paramount in most cases of emergencies. Redundancy of communication is guaranteed by the successful implementation of calling functionality and adds to the strength of the entire safety mechanism.

Operation reliability and accountability are also provided by the system through the support of alert cancellation and history maintenance. Alerts can be terminated in a controlled way to avoid needless escalation in case the user is safe, and history records allow reviewing and analyzing the incident after it has occurred. These are features that are not commonly found in non-pro complex safety applications but are needed to allow responsible deployment of the system and improvement in the future.

Despite these strengths, certain limitations remain. The system's performance is dependent on network connectivity and GPS availability, which may be constrained in remote or indoor environments. Additionally, large-scale deployment would require further evaluation of backend scalability and data security considerations. These aspects present opportunities for future enhancement.

Although these strengths exist, there are some weaknesses. The functionality of the system

requires network connectivity and GPS positioning that might be limited in the remote or in-door setting. Moreover, the scalability of backends would have to be assessed further to implement it on a large scale and data security issues. These are areas that can be improved in the future.

In general, the discussion implies that the proposed Women Safety Application based on Flutter manages to combine technological possibilities with the needs in the real world regarding the emergency. The system is proved to be practically efficient and yet simple enough, which allows implementing it in real life and developing it in the course of the research.

VII. CONCLUSION AND FUTURE WORK

In this paper, an application was described named Women Safety Application created using flutter that is used to assist in fast emergency response by means of instant SOS activation, constant GPS location tracking, live location sharing and emergency calling. The proposed system will overcome the major shortcomings of current safety solutions since it will require minimum human interaction and ensure consistent situational awareness throughout emergencies. The outcomes of the experiment proved the valid alert generation, real-time tracking, and efficient emergency lifecycle management. The system is realistic in the field of application and increases coordination of responders. The next stage of work will be aimed at enhancing network time-resistance, incorporating wearable sensors, and adding intelligent threat detection to make women- centric safety systems even stronger.

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